

Introduction to Linux and Embedded Linux

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1 Introduction

Linux is a free and open-source operating system based on the Unix operating system. It provides features such as multitasking, multi-user support, process management, memory management, networking, security, and device driver support.

Linux is widely used in servers, desktops, cloud computing, networking equipment, smartphones, and embedded systems.

2 What is Linux?

Linux consists of several major components.

- Linux Kernel
- Shell
- File System
- Device Drivers
- System Libraries
- User Applications

The Linux kernel is the core of the operating system. It manages CPU scheduling, memory allocation, file systems, hardware devices, networking, and communication between software and hardware.

3 Architecture of Linux

User Applications

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System Libraries

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Linux Kernel

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Hardware

4 What is Embedded Linux?

Embedded Linux is a customized Linux operating system designed specifically for embedded devices. It is optimized for a particular hardware platform and application.

Examples include:

- Raspberry Pi
- Industrial Controllers
- Smart TVs
- Routers
- Medical Devices
- Automotive Systems
- IoT Devices

Embedded Linux provides the advantages of Linux while allowing developers to remove unnecessary components to reduce memory usage and improve performance.

5 Linux Boot Process

The Linux boot process consists of the following stages.

1. Power On
2. Bootloader starts
3. Linux Kernel loads
4. Device Drivers initialize
5. Root File System mounts
6. Init/Systemd starts
7. User Login

6 Linux File System

Everything in Linux is represented as a file.

Some important directories are shown below.

Directory	Purpose
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/	Root directory
/bin	Basic user commands
/boot	Bootloader and kernel files
/dev	Device files
/etc	Configuration files
/home	User files
/lib	Shared libraries
/media	Removable media
/mnt	Temporary mount points
/opt	Optional software
/proc	Process information
/root	Root user home
/sys	Kernel information
/tmp	Temporary files
/usr	User applications
/var	Log and variable files

7 Linux Commands

The following commands are frequently used in Embedded Linux development.

Command	Description
pwd	Display current directory
ls	List files and folders
cd	Change directory
mkdir	Create directory
rmdir	Remove empty directory
touch	Create a file
cp	Copy files
mv	Move or rename files
rm	Delete files
cat	Display file contents
nano	Edit text files
clear	Clear terminal
history	Show previous commands
find	Search files
grep	Search text inside files
chmod	Change file permissions
ps	Display running processes
top	Monitor system processes
kill	Terminate a process
df	Display disk usage
du	Display directory size
free	Display memory usage
uname -a	Display system information
hostname	Show system hostname
ping	Test network connectivity

ifconfig / ip addr	Display network information
scp	Secure file transfer
ssh	Remote login
gcc	Compile C programs
make	Build projects
dmesg	Display kernel messages
systemctl	Manage system services
journalctl	View system logs
i2cdetect	Detect I2C devices

8 Linux System Calls

Applications communicate with the Linux kernel through system calls.

Some commonly used system calls are:

- open()
- read()
- write()
- close()
- ioctl()
- fork()
- exec()
- wait()
- mmap()
- socket()

9 Advantages of Embedded Linux

- Open source
- Reliable
- Stable
- Secure
- Portable
- Supports networking
- Supports multitasking
- Large developer community

- Excellent hardware support
- Easy software development

10 Applications

Embedded Linux is widely used in

- Automotive Electronics
- Consumer Electronics
- Robotics
- Industrial Automation
- Medical Equipment
- Smart Home Devices
- IoT Systems
- Communication Devices

11 Conclusion

Linux is a powerful operating system that provides multitasking, networking, security, and hardware support. Embedded Linux is a customized version of Linux designed for embedded devices with limited resources. Knowledge of Linux commands, file systems, system calls, and device drivers is essential for Embedded Linux development.